SFP-10GSR-85 devices, one of each of the two brands was exposed to a higher fluence of 14×10^{12} (1 MeV equivalent n)/cm^2, and a batch of ten of each of the two brands was exposed to 7×10^{12} (1 MeV equivalent n)/cm^2 fluence. All of the samples irradiated from Direktronik and FS-Europe were fully functional after irradiation.

It was seen that the threshold voltages for the FDFMA3N109 MOSFETs used in the design shifted with the exposure (figure 4) due to the Gamma component present in the reactor cavity (195 Gy). The asymmetrical profile of the flux in the cavity was used to have a perception of the correlation between the shifts of the thresholds in the MOSFETs and the exposed fluence. Four out of fifteen QUAD oscillators LTC6902IMS irradiated failed to pass the post irradiation tests, therefore the device needs to be removed from the design. The rest of the components (DAN217T146CT Schottky diodes, 540BAA100M000BBG oscillators and INA333AIDRG instrumentation amplifiers) passed the post radiation tests with no issues.

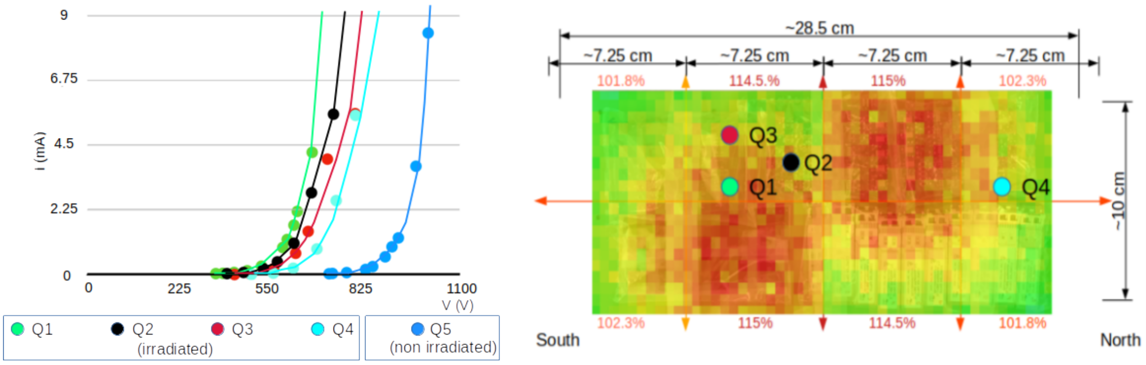

Figure 4. (Left:) Threshold shift depicted in an IV curve of four of the irradiated FDFMA3N109 MOSFETs. (Right:) Position of the irradiated MOSFETs in the flux 2D slice map of the reactor cavity.

5 Conclusions

Radiation tests were conducted to determine the suitability of each component and part of the DB6 design for meeting HL-LHC requirements. The results of the completed tests have demonstrated that the key components of the design have TID, NIEL, and SEL resistance up to the necessary radiation levels, with a few non-critical, radiation-sensitive parts still requiring attention. Additional TID and SEL testing needs to be carried out on certain SFP+ variants.

Some radiation sensitive parts of the features of the design need modification:

- The supplementary over-current protection circuit added for SEL mitigation has to be removed because of the radiation induced variations in the threshold of the FDFMA3N109 MOSFETS used in the design.
- The LTC6902IMS QUAD oscillators, previously used to de-couple the phase of the DC-DC converters, has to be removed from the design. Further research is necessary to evaluate the effect of this design modification on board noise levels.

The removal of the aforementioned MOSFETs and QUAD oscillatorswill make the design components at a board level sufficiently resistant to the expected TID, NIEL ans SEL irradiation levels.

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